

Haiqing Gao

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Technological Institute, 2145 Sheridan Road, L375, Evanston, IL, 60208

Research Interest

I'm interested in applying mathematical modeling and stochastic optimization to the design and control of non-profit operations. My current work focuses on uncovering structural properties of optimal allocation and routing policies, with the goal of enabling more effective machine learning approaches for online sequential allocations in non-profit contexts. Previously, I have conducted data-driven research in areas such as online matching platforms, preventive healthcare, and distributed learning. Broadly, my research aims to deepen the theoretical understanding of mathematical models to generate actionable managerial insights and support effective decision-making.

Education

Northwestern University

Ph.D. Candidate in Industrial Engineering and Management Science

Sep. 2022 – Current

Advised by Prof. Iravani M.R. Seyed and Prof. Sean R. Sinclair

The Chinese University of Hong Kong, Shenzhen

M. phil. in Computer Information and Engineering

Sep. 2020 – Jun. 2022

B.S. in Applied Mathematics (First Honor)

Sep. 2016 - May 2020

Publications

History-dependent or memoryless? Objective Selection for Fair Allocation and Routing in Nonprofits

Haiqing Gao, Seyed M.R. Iravani, Dawson J. Ren, Sean R. Sinclair

(Working Paper)

FedHD: Communication-efficient federated learning from hybrid data

Haiqing Gao, Songyang Ge, Tsung-hui Chang

Journal of the Franklin Institute (2023)

Presentations

History-dependent or memoryless? Objective Selection for Fair Allocation and Routing in Nonprofits

- INFORMS Annual Meeting, Public Sector OR Oct. 2025

Session Chair & Speaker

- 3rd Annual Supply Chain and Operations Management Conference (SCOM) Aug. 2025

Invited Talk for Dedicated Student Session

Fair Allocation in Nonprofit Operations: Optimal Routing Policy

- Chicago Area Society for Industrial and Applied Mathematics (SIAM) Student Conference 2025

(CASSC25) Mar. 2025

Poster & Lightning Talk

- Toyota Technological Institute at Chicago (TTIC) Workshop

Aug. 2024

Poster & Lightning Talk

Federated Q-learning in Mobile Health (mHealth) Intervention

- Reinforcement Learning Projects Presentations for Graduate Students

Nov. 2020

Poster & Lightning Talk

Fair Order Dispatch on the Online Ride-Hailing Platform

- Shenzhen Research Institute of Big Data Research Presentation

Jul. 2019

Honors

The Chinese University of Hong Kong, Shenzhen

Dean's List (Top 10%)

2016-2019

Master's List (Top 10%)

2017-2019

Academic Performance Scholarship (Top 7%)

2017

Teaching Experience

Lecturer

Northwestern University

Mar. 2024 - Apr. 2024

Two Lectures in Industrial Engineering and Management Science (IEMS) 302: Combinatorics, Conditional Probability, Continuous Distributions

The Chinese University of Hong Kong, Shenzhen

Online Teaching for High School Students in Oral English

Mar. 2020 - Apr. 2020

Teaching Assistant

Northwestern University

IEMS 365: Analytics for Social Good

Winter Quarter 2025

IEMS 303: Statistics

Fall Quarter 2024

IEMS 307: Quality Improvement by Experimental Design

Spring Quarter 2024

IEMS 302: Probability

Winter Quarter 2024

IEMS 201: Introduction to Statistics or Data-Driven Decision-Making

Fall Quarter 2023

The Chinese University of Hong Kong, Shenzhen

MATH 3253: Complex Variables

Spring Semester 2022

STA 2001: Probability and Statistics I

Spring Semester 2019

Mentorship

Online Peer Mentor for the English Language Program Foundations

Jun. 2024 – Current

The Graduate School at Northwestern University

- Held weekly meetings for 60 incoming international students on accommodating to the life in the USA
- Served as mock students to interact with candidates in teaching demonstration assessments

INFORMS Peer Mentor

Dec. 2023 - Current

INFORMS Student Chapter at Northwestern University

- Mentored two first-year IEMS Ph.D. students

TGS Diversity Peer Mentor

Dec. 2023 – Current

The Office of Diversity and Inclusion at Northwestern University

- Mentored two first-year Northwestern international Ph.D. students

Skills

Programming Languages/Packages: Python, MATLAB, R, Java, LaTeX, TensorFlow, Keras

Languages: Chinese (Native), English (Fluent)